Who to Call, When to Call:
A Data-Driven Framework for Term Deposit Marketing Success

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1. Executive Summary

This report analyses a Portuguese bank's term deposit marketing campaign to identify which customers, campaign designs, and timing decisions generate profitable growth. Using 45,211 customer records with 17 attributes from past campaigns, the study seeks to assess how sociodemographic factors, contact strategy, and call timing jointly shape subscription outcomes and financial returns.

The analysis combines segment-level profiling, logistic regression, and the Expected Monetary Value (EMV) framework. Results show that while age and job meaningfully differentiate response rates, campaign execution variables such as call duration, contact history, and month can deliver far superior predictive power. The EMV matrix was able to identify a small set of "golden" age-to-job segments with high conversion rates and strong positive EMV. For example, *"Not-Labor-Force",* including customers aged 60+, have an EMV near +$43 per contact with a ROI of above 100%. On the other hand, the model locates several "bleeding" segments where the call yields negative values.

The report recommends reallocating calls and budgets towards profitable segments, redesigning campaigns for unprofitable groups, and tightening contact rules around the most responsive periods.

2. Project Goals

Main question: How campaign impacts differ by sociodemographic group in term deposit marketing for a banking client?

- **RQ1:** How do sociodemographic factors associate with term deposit subscription propensity? *(Bivariate Analysis)*

- **RQ2:** What is the comparative predictive power of demographic versus campaign execution variables? *(Logistic Regression Model Comparison)*
- **RQ3:** Do campaign effectiveness patterns differ systematically across customer segments, requiring tailored strategies? *(Interaction Effects Analysis)*
- **RQ4:** Which customer segments merit continued investment under a financially disciplined framework, accounting for both acquisition costs and customer value? *(EMV-Based Decision Analysis)*

3. Literature Review

Term-deposit marketing is often studied as an example of how data can improve decision-making in retail banking, with the Portuguese telemarketing dataset serving as a standard empirical reference. (Moro et al., 2011). Prior research frames the task as a binary classification problem, attempting to prove that structured data-mining methods can improve campaign efficiency. However, most studies focus mainly on maximizing prediction accuracy rather than evaluating whether these predictions translate into real sustainable financial value for the bank. This present study builds on this foundation by explicitly linking segmentation and predictive modeling through an Expected Monetary Value (EMV) framework.

Research on ageing and life-cycle wealth shows that people's financial situation changes over their lifetime and this affects their demand for "safe savings products" such as term deposits (Olivera and De Mulder, 2025). Evidence across Europe shows that wealth usually rises over adulthood, peaks around retirement, and then declines slowly, especially in aging countries such as Portugal, Spain, and Italy. Older people, therefore, tend to enter deposit campaigns with more savings and a stronger need to protect their money, while younger adults are more likely to focus on day-to-day spending, borrowing (credit), and flexibility within their investments.

While many earlier banking studies include “age” as a background control variable, our research suggests that it should play a much more central role in understanding saving behavior. For this reason, age is treated as a key way of grouping customers and is divided into life-stage bands when analyzing both response and long-term value.

Within this life-cycle view, marketing research also shows that customer behavior should not be understood using demographics alone (Alves Gomes and Meisen, 2023). More effective segmentation combines who the customer is and how they behave and interact with the organization. Studies across both e-commerce and financial services show that combining demographic information such as age and occupation with behavioral information such as past contacts, product use, and channel preference leads to better targeting than using demographics on their own.

Simple demographic groupings can miss important differences in engagement and profitability within the same age or job category. This supports the age/job segmentation, where job categories are grouped into broader economic types and combined with age groups to examine whether different groups should be approached in different ways. Previous research on the Portuguese dataset also shows that campaign-related variables usually explain customer decisions better than demographics alone (Moro et al., 2011). Variables such as how long the call lasts, how often the customer is contacted, and the month of contact are often the strongest predictors of whether someone subscribes. Demographic variables such as age and job usually only add a small improvement once these campaign factors are included. This suggests that *how* and *when* the bank interacts with customers often matters more than their personal characteristics.

However, an important limitation of this approach is that it prioritizes short-term response over longer-term customer value. In this study, this issue is addressed by comparing a model based only on demographics, a model based only on campaign variables, and a combined model, using AUC to measure how well each one separates subscribers from non-subscribers.

The timing of when customers are contacted also plays an important role. Industry evidence shows that people are more open to making financial decisions at certain times of the year, especially after holidays and during periods when they review their finances (Level Agency, 2025). Campaigns that match these periods of higher attention tend to perform better, particularly when messages are relevant to the customer. This helps explain why certain months, such as March, September, October, and December, show much higher success rates. While these effects are often treated as technical “month variables” in statistical models, they likely reflect real behavioral changes in financial planning rather than random fluctuations (Moro et al., 2011; Level Agency, 2025).

Finally, research on ageing and wealth inequality shows that measuring success using response rates alone can be misleading (Olivera and De Mulder, 2025). Some groups may respond frequently but hold little money, while others may respond less but generate much higher value for the bank. Older and retired customers, for example, tend to hold higher balances, so their subscriptions are often more profitable even if their response rates are like other groups. For this reason, Expected Monetary Value (EMV) is used to assess marketing performance by combining the probability of success, the financial return, and the cost of contacting customers (EMV Guide, 2025).

By linking sociodemographic segmentation, predictive modelling, and interaction effects with the EMV framework, this study ranks age/job groups by expected profit per contact. This shifts the focus from simply identifying who is most likely to subscribe to deciding which customer groups should continue to receive expensive phone calls, which should be moved to the cheaper digital channels, and which should no longer be targeted under a financially realistic strategy.

4. Methodology

Data & Sample

The analysis utilizes a comprehensive dataset of 45,211 customer contacts from a European retail bank's term deposit campaign. The dataset captures 17 variables spanning demographics (age, occupation, education, marital status), financial position (account balance, existing products), and campaign characteristics (contact method, duration, frequency, historical outcomes).

Feature Engineering

Based on the recent academic and industry research, we constructed two derived variables:

- Age Band (Ordinal): Five-category ordinal variable (18-29, 30-39, 40-49, 50-59, 60+) enabling life-stage analysis.
- Job Group (Categorical): Four-category classification consolidating 12 occupations into economically meaningful clusters: White-Collar (management, admin, technician, services; N=26,380), Blue-Collar (manual labor, housemaid; N=10,972), Self-Employed (entrepreneurs; N=3,066), and Not-Labor-Force (retired, students, unemployed; N=4,505).

Analytical Approach

All analyses were conducted in R, using standard functions for chi-square tests, logistic regression, and marginal effects estimation.

Phase 1: Bivariate Profiling: Chi-square tests of independence assessed demographic-outcome associations.

Phase 2: Predictive Modeling: Binomial logistic regression models were estimated (Demographics Only, Campaign Only, Combined) with discrimination quantified via AUC.

- Model 1 (Demographics): Age band, job group, education, marital status

- Model 2 (Campaign): Contact type, duration, frequency, previous contacts, historical outcome
- Model 3 (Combined): All predictors from Models 1 & 2

Variance inflation factors were inspected to check for multicollinearity and were found to be within acceptable ranges indicating no severe collinearity among predictors. Model discrimination was quantified via AUC, with values >0.80 indicating excellent performance.

Phase 3: Interaction Analysis: To test whether campaign tactics exhibit differential effectiveness across segments, we estimated logistic models with interaction terms (Duration × Age Band; Frequency × Job Group). Marginal effects were visualized to support strategic interpretation.

Phase 4: Financial Decision Framework: Finally, we applied an Expected Monetary Value (EMV) analysis using both standard ($200 LTV) and balance-adjusted (Balance 5% margin) methodologies.

$Standard\ EMV\ model\ =\ Expected\ Revenue\ \ -\ Expected\ Cost$

$$Balance\ -\ Adjusted\ EMV\ Model\ =\ Expected\ Adjusted\ Revenue\ -\ Expected\ Cost$$

1. Expected Revenue: $200*Conversion Rate. The value of a success ($200 assumption) multiplied by the chance of getting it (Conversion Rate).
2. Expected Cost: $15*Average Contacts. The cost of a single call ($15 assumption) multiplied by the average effort required (Average Contacts).
3. Revenue per success: Average balance*0.2, supposing that a bank earns 2% per year on each deposit. This approximates a 2% annual margin over a 10-year horizon.
4. Expected Adjusted Revenue: Revenue per success*Conversion Rate
5. Expected Adjusted Cost: $15*Average Cost

Note: The EMV parameters act as illustrative benchmarks intended to approximate plausible banking economics rather than exact internal figures. EMV results should therefore be interpreted as relative rankings of segment profitability rather than precise profit forecasts.

5. Findings & Discussions

The Demographic Profile: Bivariate Analysis (RQ1)

Table 1: Subscription Rates by Age Band

Age Band	N	Subscribers	Conversion Rate
18–29	5,273	928	17.6%
30–39	18,089	1,913	10.6%
40–49	11,655	1,063	9.1%
50–59	8,410	785	9.3%
60+	1,784	600	33.6%

Chi-Square values (χ^2 = 1151.16, $p < 0.001$)

Table 1 represents the subscription rate across age groups. Customers over the age of 60 exhibit by far the highest conversion rate (33.6%), indicating a strong affinity for term-deposit products among older individuals. However, the 30-39 age band has the largest number of subscribers (1913), due to its size, but holds a relatively small conversion rate (10.6%). Young adults (18-29) convert at a rate of 17.6%, while people from 40-49 and 50-59 years of age convert at similar, low response rates, being 9.1% and 9.3%, respectively. The Chi-square test (χ^2>p) indicates that these differences are statistically significant, meaning that subscription likelihood varies significantly by age. These patterns suggest that while older individuals remain the most responsive, younger individuals (30-39) have the potential to add long-term customer value and support future campaign performance.

Table 2: Subscription Rates by Job Group

Job Group	N	Subscribers	Conversion Rate
Not-Labor-Force	4,505	987	21.9%

White-Collar	26,380	3,141	11.9%
Self-Employed	3,066	310	10.1%
Blue-Collar	10,972	817	7.4%

Chi-square (χ^2 = 655.31, $p < 0.001$)

Table 2 summarizes how term-deposit subscription rates vary across the four job categories. Individuals who are not in the labor force (retired, students, unemployed) display the highest conversion rate (21.9%), despite representing a smaller proportion of total contracts. In contrast, white-collar customers generate the largest absolute number of subscribers (3,141) due to their much larger group size, but their conversion rate is lower (11.9%). Self-employed and blue-collar segments show comparatively weaker responses, with conversion rates of 10.1% and 7.4%, respectively. The Chi-square test (□^{2}>p) confirms that these differences are statistically significant, indicating that the differences in job groups are significant regarding the subscription outcomes.

Overall, the variables in both bivariant analysis models are statistically significant, and the analysis shows that both demographic characteristics and campaign design play important roles in determining term-deposit subscription success.

Execution Over Targeting: The Primacy of Campaign Variables (RQ2)

Table 3: Logistic Regression Model Performance Comparison (AUC)

Model	AUC
Model 1: Demographics (Age groups)	0.642
Model 2: Campaign Execution	0.899
Model 3: Combined	0.906

The comparison of the explanation power of the three logistic regression models based on their cross-validation AUC shows a clear result: campaign execution explains subscription outcomes much more strongly than demographic characteristics. Model 1, which includes only demographic variables, achieves an AUC of 0.642. This level of performance indicates that demographics have some predictive value, but they are not strong enough to reliably separate customers who will subscribe from those who will not. In other words, knowing who the customer is helps only to a limited extent.

Model 2 changes the picture entirely. When only campaign-related variables are used, the AUC rises sharply to 0.899. This represents a very high level of predictive accuracy and shows that campaign design and customer interaction characteristics (such as duration of the call, number of previous contacts, and the outcome of earlier campaigns) drive most of the explanatory power. How the bank approaches the customer is far more important than the customer's socio-demographic background.

Model 3 combines both variable groups and reaches an AUC of 0.906. Although this is the best-performing model, the improvement over Model 2 is minimal. This suggests that demographic characteristics add only marginal information once campaign execution variables are included. A likely explanation is that campaign variables capture behavioral signals, what the customer did during and before the contact, while demographic variables reflect static characteristics. Behavioral indicators tend to be much stronger predictors in marketing analytics because they track engagement and prior interest. Moreover, historical campaign outcomes are often among the best predictors of future responses, and this dataset includes several such variables (for example, "previous" and "outcome"), which naturally dominate the model.

Overall, the results point to a clear conclusion: improving the way campaigns are executed has a much larger impact on performance than changing the targeting strategy based on demographic profiles.

Table 4: Logistic regression coefficients: Model 3

Predictor	Coefficient	Odds Ratio	p-value
Top 5 Seasonal Timing (Ref: January)			
March	2.75	15.60	< 0.001
October	2.07	7.94	< 0.001
September	2.01	7.45	< 0.001
December	1.88	6.58	< 0.001
June	1.62	5.05	< 0.001
Campaign History (Ref: Failure)			
Previous Success	2.38	10.82	< 0.001
Other Outcome	0.23	1.26	0.008
Job Group (Ref: White-Collar)			
Not-Labor-Force	0.23	1.26	< 0.001
Self-Employed	-0.18	0.84	0.021
Blue-Collar	-0.24	0.79	< 0.001
Age Band (Ref: 18-29)			
60+	0.46	1.59	< 0.001
30-39	-0.42	0.66	< 0.001
40-49	-0.42	0.66	< 0.001
50-59	-0.43	0.65	< 0.001
Education (Ref: Secondary)			
Tertiary	0.24	1.27	< 0.001
Unknown	0.18	1.20	0.045

Primary	-0.19	0.83	0.004
Marital Status (Ref: Single)			
Married	-0.28	0.76	< 0.001
Divorced	-0.09	0.92	0.200
Contact Type (Ref: Cellular)			
Telephone	-0.18	0.84	0.020
Unknown	-1.65	0.19	< 0.001
Continuous Variables			
Duration (continuous)	0.00	1.00	< 0.001
Campaign Frequency (continuous)	-0.08	0.92	< 0.001

According to Table 4, seasonal timing and campaign history stand out as the most influential predictors in the subscription probability of subscribing to a term. Looking at seasonal timing, March, October, September, December, and June show very strong positive effects, with odds ratios ranging from 15.60 to 5.60 compared with January. This means customers contacted in these months are 15 to 5 times more likely to respond compared to January. For instance, after the New Year, people return to their usual routines and may subscribe to a term deposit in March. Similarly, after summer, customers resume structured planning and financial activities, such as subscribing to term deposits. Turning to campaign history, customers who succeeded in a previous campaign are over 10 times more likely to respond again, confirming that past behavior is a strong predictor of future behavior. This may be due to familiarity with the brand, established trust, or recurring needs and motivations.

Other factors such as age band, job group, and education levels have moderate effects in the power prediction. The age band shows that elderly people (60+) are the only group with more potential to have a deposit compared with the youngest group. Employment status also matters; non-working individuals have more odds to subscribe to a term compared with white-collars groups probably based on their economic necessities. Education level shows that higher-educated customers tend to have more chances to subscribe than a client with just secondary school term ends.

Reference levels for categorical variables were manually specified to enhance interpretability: Age Band (18-29), Job Group (White-Collar), Education (Secondary), Marital Status (Single), Contact Type (Cellular), Month (January), Prior Outcome (Failure). These references represent either the most common category or a logical baseline (e.g., January for seasonal comparison).

Finally, most predictors in Table 4 are statistically significant, except for one notable case. The marital status category for divorced customers ($p = 0.200$) does not seem to affect the likelihood of a response. Including this variable still helps provide a complete overview of all factors in the analysis, even if its individual influence is limited.

Tailoring the Approach: Interaction Effects Across Segments (RQ3)

While demographics lack standalone predictive power, they can **moderate** the effectiveness of campaign tactics. We tested two important interactions:

Interaction A - Duration × Age Band:

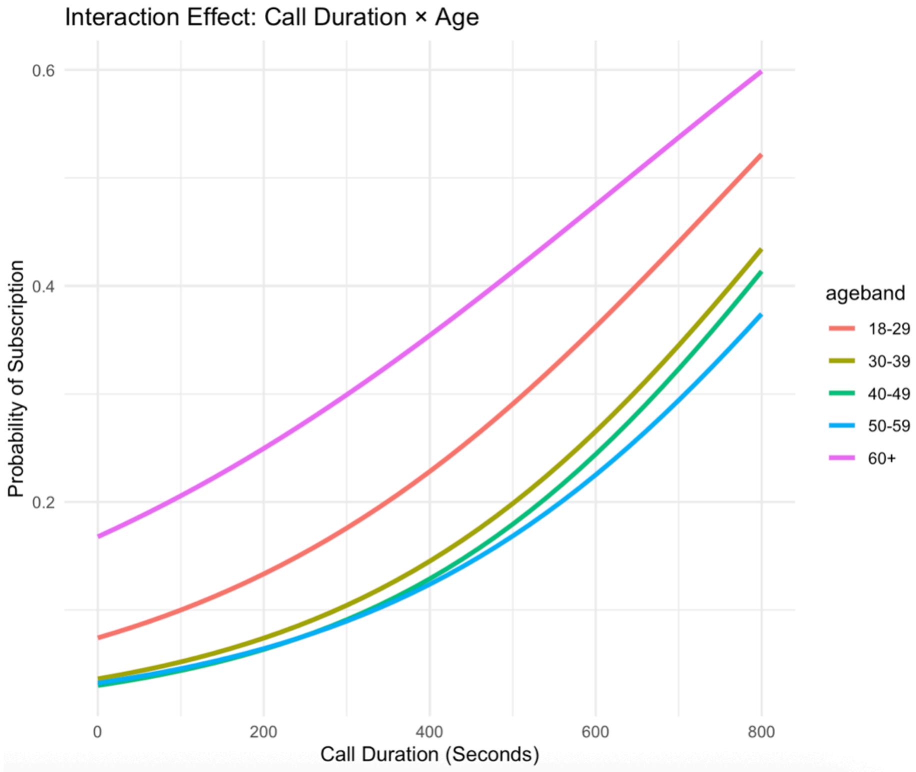

Figure 1: Interaction Effect A - Call Duration × Age

Figure 1 shows that across all age groups, longer calls increase the probability of subscription, though the magnitude of this effect varies substantially. Individuals over 60 years of age react the strongest: the slope is approximately +0.08 per 100 seconds, indicating that seniors respond particularly well to longer, more consultative, and relationship-building engagements. In contrast, younger cohorts display flatter curves, with benefits vanishing after roughly 300 seconds, suggesting diminishing returns from extended conversations, as working professionals (ages 30-59) prefer more efficiency-driven calls.

Frequency × Job Group: Blue-Collar Fatigue:

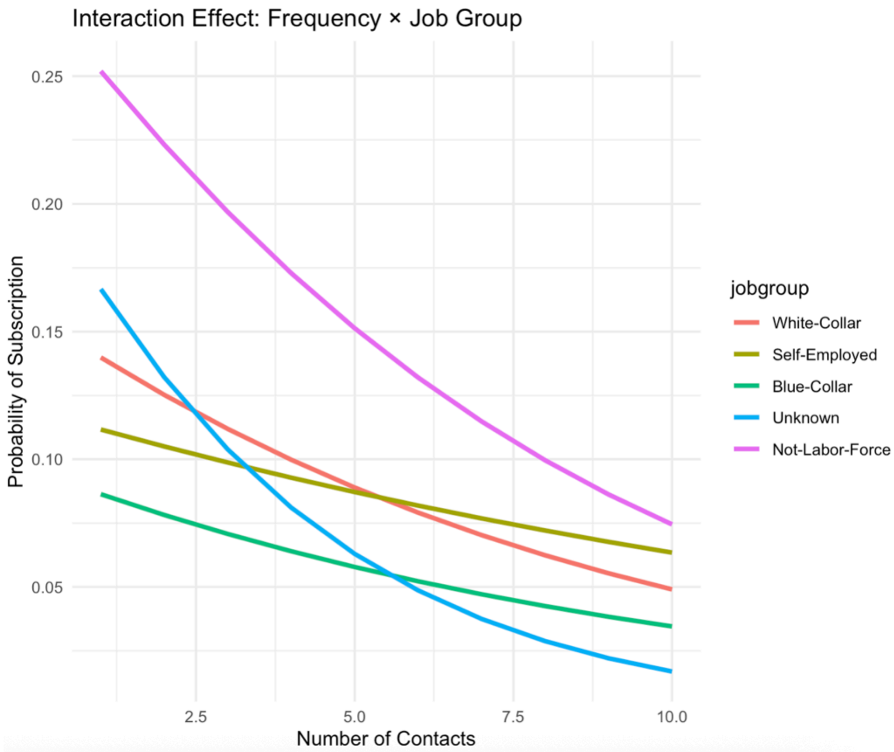

Figure 2: Interaction B - Frequency × Job Group

Figure 2 shows how repeated contact affects subscription likelihood across job segments. Across all groups, additional calls reduce the probability of subscribing, indicating diminishing returns from repeated outreach. The group of individuals who are not part of the labor force starts with the highest baseline probability (roughly 25%) but experiences a steady decline as contact frequency increases, suggesting that even more receptive groups become less responsive when approached too often. White-collar and self-employed customers move in parallel, showing moderate responsiveness to repeated calls, while blue-collar and “unknown” groups exhibit the steepest drop-offs, with subscription probabilities falling sharply after a few contacts. These patterns indicate that over-contacting is mostly counterproductive for working-age and lower-engagement groups, reinforcing the importance of stricter contact-frequency limits.

The Financial Reality: Where We Lose Money (RQ4):

Table 5: Strategic Decision Matrix (Ranked by EMV)

Job Group	Age Band	N	Conv. Rate	Avg. Contacts	Avg. Balance	Cost	EMV (Standard)	EMV (Adjusted)
Not-Labor-Force	60+	1,108	37.5%	2.11	$2,500.87	$84.48	+$43.37	+$15.23
Not-Labor-Force	18-29	880	28.4%	2.21	$1,304.87	$116.58	+$23.70	-$14.58
White-Collar	60+	382	29.8%	2.63	$2,353.59	$132.11	+$20.26	-$4.31
Not-Labor-Force	30-39	700	16.4%	2.56	$1,425.42	$233.48	-$5.50	-$26.65
Blue-Collar	60+	199	20.1%	2.51	$1,808.21	$187.12	+$2.59	-$19.44
Self-Employed	18-29	277	17.3%	2.39	$930.81	$206.56	-$1.14	-$27.73
White-Collar	18-29	2,912	16.8%	2.50	$1,008.03	$223.01	-$3.86	-$28.98
White-Collar	30-39	11,908	11.6%	3.17	$1,255.49	$367.61	-$19.41	-$35.30

Note: Top 8 groups sorted by the EMV (Standard). N = Segment size. Conv. Rate = Proportion subscribing. Avg. Contacts = Mean contact attempts per customer. Avg. Balance = Mean account balance for segment. Acq. Cost = (Avg Contacts / Conv Rate) × $15. EMV (Standard) = (Conv Rate × $200) - (Avg Contacts × $15). EMV (Adjusted) = (Conv Rate × Avg Balance × 0.05) - (Avg Contacts × $15).

Table 5 reveals three segments-Not-Labor-Force 60+ (Rank 1), Not-Labor-Force 18-29 (Rank 2), and White-Collar 60+ (Rank 3) that appear profitable under the standard $200 LTV assumption, collectively representing 2,070 customer contacts (2.3% of total volume) with positive estimated returns. However, when adjusted for actual account balances, the landscape shifts dramatically. Not-Labor-Force 18-29 (Rank 2), despite showing +$23.70 Standard EMV, becomes a value destroyer at -$14.58 Adjusted EMV because their smaller average balance ($1305) cannot justify the $116.58 acquisition cost. Similarly, White-Collar 60+ (Rank 3) swings from +$20.26 to -$4.31, revealing that despite high conversion (29.8%), higher acquisition costs ($132.11) relative to their balance-derived value make them unprofitable under realistic margin assumptions.

Only the *"Not-Labor-Force"* 60+ segment survives both tests. This segment, representing just 1,108 contacts (2.5% of volume), should become the exclusive focus of the expensive phone channel. All other segments-particularly the large White-Collar 30-39 cohort (Rank

8, N=11908) with -$35.30 Adjusted EMV-should immediately migrate to digital channels or be deprioritized.

6. Recommendations

Recommendation	Segment	Rationale	Implementation
Refocus calls on high EMV seniors	"Not-In-Labor-Force", 60+	It is the only consistently profitable segment that shows strong responsiveness to longer, consultative calls	Design a "premium senior" calling playbook. Must include: a relationship oriented script, 8-12 min call target, explicit needs assessment, reassurance on safety of funds. Allocate more experienced agents. Priority in dialler queues.
Adopt EMV as the primary targeting KPI	Campaign planning and targeting decisions	Response-based optimisation over-weighs both the high and low value segments and under-weighs smaller but profitable niches	For all future marketing campaigns rank segments by EMV and not the conversion rate. Integrate EMV dashboards into campaign design reviews.
Move negative value segments to "digital first" journeys	White-collar 18–39 and 30–39, Self-Employed 18–39	Groups that generate acceptable or high response rates but deliver negative adjusted EMV due to their high acquisition costs	Replace repeat phone calls with email, in-app and SMS journeys that promote online self-service deposit offers. Retain a "call back on request" option.
Establish a quarterly "profitability	All segments and channels	EMV rankings depend on assumptions and may shift as the macro environment and	Re-estimate conversion rates, balances and average call costs every quarter and

refresh" cycle		customer behaviour change	recompute EMV by age/job segment. Adjust these priorities based on updated rankings.
Focus your campaign with particular insights on peak months	Implementation strategy	March, October and September show the highest levels of probabilities of a success subscription term.	Run 1 or 2 pilot campaigns applying messages focused on the necessities of each period of time.

7. Conclusion

The project examined how a Portuguese bank's term-deposit telemarketing campaign performs across sociodemographic groups and which customers generate genuine financial value once marketing costs and customer balances are considered. Using 45,211 contacts and four phase frameworks to reinforce a central theme from the literature: while demographics shape baseline demand, campaign execution and financial discipline ultimately determine profitability.

In line with the life cycle wealth theory, older and "*Not-In-The-Labor-Force*" exhibited the highest conversion rates, reflecting stronger balance positions and greater preference for more secure investments. By contract, the large 30-39 segment had generated many subscribers mainly due to scale rather than a strong individual likelihood of conversion. Consistent with prior findings in the Portuguese dataset, the logistic models show that the campaign variables (e.g. call duration, contact history) explain most of the variation in customer response, with demographic characteristics adding only limed value once the behavioral factors are controlled for. This further demonstrates that campaign tactics are not universally effective. So, while longer, consultative calls significantly improve outcomes for older people, they simultaneously erode responsiveness amongst working age and blue-collar groups. This confirms that a uniform contact strategy is inefficient and that campaign design must be segment specific.

The EMV framework converts these behavioral patterns into explicit financial decision rules. Only in the "*Not-In-The-Labor-Force*" 60+ segment remains consistently profitable under both standard and balance adjusted EMV, while several large white-collar groups were value destroying despite having acceptable response rates. The findings directly support the literature's warning that response optimized targeting can misallocate marketing budgets. High predicted conversion alone cannot guarantee commercial viability once acquisition costs and customer balance are jointly considered.

Strategically, the bank should concentrate outbound phone calls on the highest EMV age/job segments particularly in the *"Not-In-Labor-Force",* where longer calls are justified by both higher response and stronger balances. Large but low EMV groups, such as white collar, should be migrated to lower cost digital channels. Similarly, call frequency and duration for working age and blue collar segment should be capped. The EMV calculations should be refreshed regularly to maintain aligned with current profitability rather than being just a single campaign snapshot.

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